# Summer Analytics 2026 — Mini-Hackathon Week 2

E-Commerce Conversion Prediction: Methodology Report

## 1. Problem & Data

The task is binary classification of the *Converted* target (1 = converted, 0 = not) using 12 anonymized user-level features (demographics, behaviour, traffic source, and campaign attributes). train.csv has 10,000 labeled records and public_test.csv has 3,000 additional labeled records used for validation. private_test.csv (3,000 records) requires predictions. The target is moderately imbalanced (~31% converted / 69% not converted).

## 2. Exploratory Data Analysis

Age, Income, and Time_On_Site contain missing values (roughly 10–18% of rows). Device_Type (Mobile/Desktop/Tablet) and Traffic_Source (Organic/Paid Ads/Social Media/Email/Referral) are categorical with no missing values. Conversion rates vary modestly across these categories (e.g., Referral and Email traffic convert slightly more than Paid Ads), and no extreme outliers were found that required removal.

## 3. Preprocessing

Missing numeric values (Age, Income, Time_On_Site) were imputed using the median. Categorical features were imputed with the most frequent value (no missing values were actually present) and one-hot encoded with unknown-category handling. User_ID was excluded from the feature set and retained only to construct the submission file.

## 4. Feature Engineering

Three derived ratio features were added to capture engagement intensity: **Pages_per_min** (Pages_Viewed / (Time_On_Site + 1)), **Products_per_page** (Products_Viewed / (Pages_Viewed + 1)), and **Income_per_age** (Income / (Age + 1)). These help the model capture browsing efficiency and purchasing-power signals beyond the raw counts.

## 5. Modeling

Logistic Regression, Random Forest, and LightGBM classifiers were compared using an 80/20 stratified train/validation split, all with class-balanced weighting to address the ~31/69 class imbalance. Random Forest (300 trees, max_depth=8, class_weight='balanced') gave the best and most stable validation F1 (≈0.567) and was selected as the final model. Threshold tuning on probability outputs was tested but the default 0.5 threshold performed best for this model.

## 6. Final Training & Prediction

For the final submission, the Random Forest model was retrained on the combined train.csv + public_test.csv data (13,000 labeled records) using identical preprocessing and feature engineering, then used to predict Converted for all 3,000 records in private_test.csv. Validation F1 on the held-out split was 0.567; F1 on public_test.csv (using the train-only model) was 0.52.

## 7. Deliverables

submission.csv (User_ID, Converted for all private_test.csv records), notebook.ipynb (reproducible end-to-end pipeline), and this one-page report.